

# Lectures on Partial Extrinsic Curvature

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## Introduction

The Theory of Partial Extrinsic Curvature (PXC) is a geometric toy theory of gravity based on the concept of extrinsic curvature. In PXC, we have two coordinate systems; the flat container space and the warped embedded space. The container space is unobservable, but the “embedding” makes up all of space as we experience it, like the material of space itself.

Lecture 1 will be discussing the basics of the embedding and it’s kinematics, or the pieces of it relating to motion without discussing where the motion comes from. We’ll also do a proof-of-concept run to reproduce the Schwarzschild line element and will use it to get Newtonian gravity from the embedding’s fictitious forces.

Lecture 2 will discuss topics on PXC, including the nature of a complex embedding and partial operations. This lecture is mainly background underlying PXC and can be safely skipped for lecture 3.

Lecture 3 is currently under active research, but will discuss the field equations of the embedding.

Note that this is a theory in progress- more will be added as my research continues.

## Lecture 1: Embedding Kinematics & Introducing Relativity

### The Lavelle Relations

To start, we’ll define the zeroth-order “Lavelle relation:”

$$|dx\rangle = \phi |dC\rangle \quad \phi = \frac{\partial |x\rangle}{\partial \langle C|}$$

For now, we can imagine  $|x\rangle$  and  $\langle C|$  to be real; later, we’ll see that this won’t be the case in practice. Thus,  $\phi$  is the Jacobian of the map  $\mathbf{x}(\mathbf{C})$ , and  $\phi |dC\rangle$  implies a summation in the rows of  $\phi$  with the column vector  $|dC\rangle$ . This relation then tells us that dimensions are mixed when transformed into embedding

coordinates; every component of  $|dx\rangle$  does not depend only on it's counterpart in  $|dC\rangle$  but instead on contributions from every direction weighted by  $\phi$ .

Defining that  $x_4 = t$  is the experienced time in the embedding, we can divide through to get the first-order Lavelle relation:

$$|\dot{x}\rangle = \phi |\dot{C}\rangle$$

Differentiating once gives the second-order Lavelle relation:

$$|\ddot{x}\rangle = \frac{\partial(\phi |\dot{C}\rangle)}{\partial t} = \frac{\partial\phi}{\partial t} |\dot{C}\rangle + \phi |\ddot{C}\rangle$$

We can chain rule the  $\partial\phi/\partial t$  derivative:

$$\frac{\partial\phi}{\partial t} = \frac{\partial\phi}{\partial\langle C|} |\dot{C}\rangle \implies |\ddot{x}\rangle = \psi |\dot{C}\rangle |\dot{C}\rangle + \phi |\ddot{C}\rangle \quad \psi = \frac{\partial|x\rangle}{\partial\langle C| \partial\langle C|}$$

This is the Lavelle relation at second order. We have two terms to discuss here:

1.  $\psi |\dot{C}\rangle |\dot{C}\rangle$  is a fictitious force sourced by the curvature of the embedding. Intuitively, we can imagine a straight line on a curved graph appearing to diverge from straight in graph coordinates. An embedding with no second order change will not generate this term; curvature, second order change, is required to create it.
2.  $\phi |\ddot{C}\rangle$  is the transformation of incoming acceleration into embedded coordinates, using the same method the first order Lavelle relation uses to convert incoming velocities into embedding coordinates. Again, like the zeroth order Lavelle relation, accelerations are mixed as their transformed.

Thus, the final set of Lavelle relations read:

$$|dx\rangle = \phi |dC\rangle \quad |\dot{x}\rangle = \phi |\dot{C}\rangle \quad |\ddot{x}\rangle = \psi |\dot{C}\rangle |\dot{C}\rangle + \phi |\ddot{C}\rangle$$

### The Schwarzschild Map

Next, we'll be matching the Schwarzschild space and time elements, which are:

$$\langle dx|dx\rangle_s = \left(1 + \frac{r_s}{4r_C}\right)^4 \langle dC|dC\rangle_s \quad dt^2 = \left(\frac{1 - r_s/4r_C}{1 + r_s/4r_C}\right)^2 dT^2$$

where  $T = C_4$  is the container time coordinate,  $r_C = \sqrt{\langle C|C\rangle_s}$ , and a subscript  $s$  on a bracket or tensor implies that a summation or indexing runs over spatial indices only. We'll define that  $1 + r_s/4r_C = A$  and  $1 - r_s/4r_C = B$ , so that this becomes:

$$\langle dx|dx\rangle_s = A^4 \langle dC|dC\rangle_s \quad dt^2 = \left(\frac{B}{A}\right)^2 dT^2$$

We can expand the LHS of the spatial equation with the Lavelle relation at zeroth order:

$$\langle dx|dx\rangle_s = \langle dC|\phi^\dagger\phi|dC\rangle = A^4 \langle dC|dC\rangle_s$$

This implies that that  $\phi^\dagger\phi$  needs to go as:

$$\phi_s^\dagger\phi_s = A^4 \cdot I$$

And for the time element:

$$dt = \frac{\partial t}{\partial T^*} dT = \phi_{44} dT \implies dt^2 = \phi_{44}^2 dT^2 = \left(\frac{B}{A}\right)^2 dT^2 \implies \phi_{44}^2 = \left(\frac{B}{A}\right)^2$$

$$\phi_{44} = \frac{B}{A} \implies x_4 = \int dT \frac{B}{A} = \int dT \frac{1 - r_s/4r_C}{1 + r_s/4r_C}$$

For the spatial part, we'll try a radial ansatz  $|x\rangle_s = \lambda(r_C) |C\rangle_s$ :

$$|x\rangle_s = \lambda |C\rangle_s \implies \phi_s = \frac{\partial(\lambda |C\rangle_s)}{\partial \langle C|_s} = |C\rangle_s \frac{\partial \lambda}{\partial \langle C|_s} + \lambda \frac{\partial |C\rangle_s}{\partial \langle C|_s}$$

Using the chain rule:

$$\phi_s = |C\rangle_s \lambda' \frac{\partial r_C}{\partial \langle C|_s} + \lambda I = \frac{|C\rangle_s \langle C|_s \lambda'}{r_C} + \lambda I$$

Now taking  $\phi_s^\dagger\phi_s$ :

$$\phi_s^\dagger\phi_s = \left( \frac{|C\rangle_s \langle C|_s \lambda'^*}{r_C} + \lambda^* I \right) \left( \frac{|C\rangle_s \langle C|_s \lambda'}{r_C} + \lambda I \right)$$

Expanding:

$$\phi_s^\dagger\phi_s = \frac{|\lambda'|^2 |C\rangle_s \langle C|_s \langle C|_s}{r_C^2} + \frac{\lambda^* \lambda' |C\rangle_s \langle C|_s}{r_C} + \frac{\lambda'^* \lambda |C\rangle_s \langle C|_s}{r_C} + |\lambda|^2 I$$

$\langle C|C\rangle_s = r_C^2$ , so:

$$\phi_s^\dagger\phi_s = |C\rangle_s \langle C|_s \left( |\lambda'|^2 + \frac{\lambda^* \lambda' + \lambda'^* \lambda}{r_C} \right) + |\lambda|^2 I$$

$\lambda^* \lambda' + \lambda'^* \lambda = (|\lambda|^2)'$ :

$$\phi_s^\dagger\phi_s = |C\rangle_s \langle C|_s \left( |\lambda'|^2 + \frac{(|\lambda|^2)'}{r_C} \right) + |\lambda|^2 I$$

This acting on  $d\mathbf{C}$  needs to give a factor of  $A^4$ , so forcing  $|\lambda|^2 = A^4$ , we have  $|\lambda| = A^2$  and:

$$|\lambda'|^2 + \frac{(|\lambda|^2)'}{r_C} = 0$$

Using our leftover degree of freedom, we can write that  $\lambda = e^{i\theta}|\lambda|$  for real  $\theta$ ; then:

$$\begin{aligned}\lambda' &= \frac{\partial(e^{i\theta}|\lambda|)}{\partial r_C} = \frac{\partial e^{i\theta}}{\partial r_C}|\lambda| + e^{i\theta}\frac{\partial|\lambda|}{\partial r_C} \\ \frac{\partial e^{i\theta}}{\partial r_C} &= e^{i\theta}i\theta' \quad \frac{\partial|\lambda|}{\partial r_C} = \frac{\partial A^2}{\partial r_C} = 2A\frac{\partial A}{\partial r_C} = 2A\frac{\partial}{\partial r_C}\left(1 + \frac{r_s}{4r_C}\right) = \frac{Ar_s}{2}\frac{\partial}{\partial r_C}\left(\frac{1}{r_C}\right) = -\frac{Ar_s}{2r_C^2} \\ \lambda' &= e^{i\theta}\left(i\theta'|\lambda| - \frac{Ar_s}{2r_C^2}\right) \Rightarrow |\lambda'|^2 = \theta'^2|\lambda|^2 + \frac{A^2r_s^2}{4r_C^4}\end{aligned}$$

Next,  $(|\lambda|^2)'$ :

$$|\lambda|^2 = A^4 \Rightarrow (|\lambda|^2)' = 4A^3\frac{\partial A}{\partial r_C}$$

We computed this earlier:

$$(|\lambda|^2)' = -\frac{A^3r_s}{r_C^2}$$

So:

$$|\lambda'|^2 + \frac{(|\lambda|^2)'}{r_C} = \theta'^2 A^4 + \frac{A^2r_s^2}{4r_C^4} - \frac{A^3r_s}{r_C^3} = 0$$

Rearranging for  $\theta'$ :

$$\theta'^2 = \frac{1}{A^4}\left(\frac{A^3r_s}{r_C^3} - \frac{A^2r_s^2}{4r_C^4}\right) = \frac{1}{A^2r_C^3}\left(Ar_s - \frac{r_s^2}{4r_C}\right)$$

We can rewrite this as:

$$\theta'^2 = \frac{r_s}{A^2r_C^3}\frac{4r_CA - r_s}{4r_C} \quad 4r_CA - r_s = 4r_C\left(1 + \frac{r_s}{4r_C}\right) - r_s = 4r_C + r_s - r_s = 4r_C$$

$$\theta'^2 = \frac{r_s}{A^2r_C^3} \Rightarrow \theta' = \frac{r_s^{1/2}}{Ar_C^{3/2}}$$

$$\theta = r_s^{1/2} \int \frac{dr_C}{(1 + r_s/4r_C)r_C^{3/2}}$$

We can rearrange with  $u = r_C^{1/2}$  for the standard arctangent integral:

$$\theta = r_s^{1/2} \int \frac{2u du}{(1 + r_s/4u^2)u^3} = 2r_s^{1/2} \int \frac{du}{u^2 + r_s/4}$$

$$\theta = \frac{2r_s^{1/2}}{r_s^{1/2}/2} \arctan\left(\frac{2r_C^{1/2}}{r_s^{1/2}}\right) + D = 4 \arctan\sqrt{\frac{4r_C}{r_s}} + D$$

for some constant of integration  $D$ . Thus:

$$\lambda = \left(1 + \frac{r_s}{4r_C}\right)^2 e^{4i \arctan\sqrt{4r_C/r_s} + iD}$$

This map, the Schwarzschild map, isn't one we'll actually use in this lecture; later, we'll have it as a check to verify our field equations once we construct them. In the meantime we can just compute acceleration, using  $\psi$ , the derivative of  $\phi$ :

$$\psi = \frac{\partial}{\partial \langle C|_s} \left( \frac{|C\rangle_s \langle C|_s \lambda'}{r_C} + \lambda I \right)$$

$$\psi = \frac{\partial |C\rangle_s}{\partial \langle C|_s} \frac{\langle C|_s \lambda'}{r_C} + |C\rangle_s \frac{\partial \langle C|_s}{\partial \langle C|_s} \frac{\lambda'}{r_C} + |C\rangle_s \langle C|_s \frac{\partial \lambda'}{\partial \langle C|_s} \frac{1}{r_C} + |C\rangle_s \langle C|_s \lambda' \frac{\partial}{\partial \langle C|_s} \left( \frac{1}{r_C} \right) + I \frac{\partial \lambda}{\partial \langle C|_s}$$

Naming  $|C\rangle_s \langle C|_s$  the outer product  $O$ :

$$\psi = \frac{I \langle C|_s \lambda'}{r_C} + \frac{|C\rangle_s I \lambda'}{r_C} + \frac{O \langle C|_s \lambda''}{r_C^2} - \frac{O \langle C|_s \lambda'}{r_C^3} + \frac{I \langle C|_s \lambda'}{r_C}$$

Factoring and combining:

$$\psi = \frac{2I \langle C|_s \lambda'}{r_C} + \frac{|C\rangle_s I \lambda'}{r_C} + \frac{O \langle C|_s}{r_C^2} \left( \lambda'' - \frac{\lambda'}{r_C} \right)$$

Contracting with  $|\dot{C}\rangle_s$ :

$$\psi |\dot{C}\rangle_s = \frac{2 \langle C|\dot{C}\rangle_s |\dot{C}\rangle_s \lambda'}{r_C} + \frac{\langle \dot{C}|\dot{C}\rangle_s |C\rangle_s \lambda'}{r_C} + \frac{\langle C|\dot{C}\rangle_s |\dot{C}\rangle_s}{r_C^2} \left( \lambda'' - \frac{\lambda'}{r_C} \right)$$

Recall  $\lambda'$ :

$$\lambda' = e^{i\theta} \left( i\theta' |\lambda| - \frac{A r_s}{2r_C^2} \right)$$

We know that  $\theta = 4 \arctan \sqrt{4r_C/r_s} + D$  approaches  $2\pi + D$  rapidly for small  $r_s$  or large  $r_C$ ; thus  $\theta' \approx 0$  and we can say, to first order in  $r_s$ :

$$\lambda' \approx -e^{2i\pi+iD} \left( \frac{r_s}{2r_C^2} \left( 1 + \frac{r_s}{4r_C} \right) \right) \approx -\frac{e^{iD} r_s}{2r_C^2}$$

Taking the middle term with complex velocity normalization  $\langle \dot{C}|\dot{C}\rangle_s = c^2$ :

$$\frac{\langle \dot{C}|\dot{C}\rangle_s |C\rangle_s \lambda'}{r_C} = c^2 |\hat{C}\rangle_s \left( -\frac{e^{iD} r_s}{2r_C^2} \right) = -e^{iD} \frac{GM |\hat{C}\rangle_s}{r_C^2}$$

Taking the real part for the observable piece (discussed later in lecture 2):

$$\text{Re}(|\ddot{x}\rangle) \supset -\cos D \cdot \frac{GM \text{Re}(|\hat{C}\rangle_s)}{r_C^2}$$

As for  $D$ , if we require that  $|x\rangle$  match  $|C\rangle$  at infinity, we get:

$$|x\rangle = \lambda |C\rangle \implies \lim_{r_C \rightarrow \infty} \lambda = 1 \implies e^{iD} \rightarrow 1 \implies D = 2\pi n \quad n \in \mathbb{Z}$$

a definition under which the cosine is always 1:

$$\text{Re}(|\ddot{x}\rangle) \supset -\frac{GM \text{Re}(|\hat{C}\rangle_s)}{r_C^2}$$

To get  $r_C$  in terms of  $r$ , the observed embedding coordinate, we'll contract  $|x\rangle_s$  with  $\langle x|_s$ :

$$\langle x|x\rangle_s = r^2 = |\lambda|^2 \langle C|C\rangle_s = |\lambda|^2 r_C^2 \implies r = |\lambda| r_C = r = \left(1 + \frac{r_s}{4r_C}\right)^2 r_C$$

$$r = r_C + \frac{r_s^2}{8r_C} + \frac{2r_s}{4} \approx r_C + \frac{r_s}{2}$$

to first order in  $r_s$ . so:

$$r_C = r - \frac{r_s}{2} \implies \text{Re}(|\ddot{x}\rangle) \supset -\frac{GM \text{Re}(|\hat{C}\rangle_s)}{(r - r_s/2)^2}$$

which is Newtonian gravity exactly up to a softening parameter  $r_s/2 = GM/c^2$ .

## Lecture 2: Topics on PXC I

We've shown that PXC can successfully reproduce the Schwarzschild elements and Newtonian gravity, so before we continue to full Relativity, we should discuss the nature of what we've built so far.

The first topic we'll discuss is called the partial operation. Recall the Schwarzschild  $\phi$ :

$$\phi_s = \frac{|C\rangle_s \langle C|_s \lambda'}{r_C} + \lambda I$$

Now, if we had used  $\phi^T \phi$  instead of  $\phi^\dagger \phi$  (transpose instead of conjugate transpose), supposing that a  $\langle v|$  represents a transpose only and that all  $\langle u|v\rangle$  are dot products,  $\phi_s^T \phi_s$  would go as:

$$\begin{aligned} \phi_s^T \phi_s &= \left( \frac{|C\rangle_s \langle C|_s \lambda'}{r_C} + \lambda I \right) \left( \frac{|C\rangle_s \langle C|_s \lambda'}{r_C} + \lambda I \right) \\ \phi_s^T \phi_s &= \frac{\lambda'^2 |C\rangle_s \langle C|_s \langle C|_s \langle C|_s}{r_C^2} + \frac{\lambda \lambda' |C\rangle_s \langle C|_s}{r_C} + \frac{\lambda' \lambda |C\rangle_s \langle C|_s}{r_C} + \lambda^2 I \\ \phi_s^T \phi_s &= |C\rangle_s \langle C|_s \left( \lambda'^2 + \frac{2\lambda \lambda'}{r_C} \right) + \lambda^2 I \end{aligned}$$

So:

$$\lambda^2 = A^4 \implies \lambda = A^2$$

Then the other terms must vanish:

$$\lambda'^2 + \frac{2\lambda\lambda'}{r_C} = 0 \implies \lambda' + \frac{2A^2}{r_C} = 0$$

Note that if  $\lambda = A^2$  it must be purely real, so  $\lambda'$  is just  $\partial A^2 / \partial r_C$ :

$$\begin{aligned} \lambda' &= 2A \frac{\partial A}{\partial r_C} = \frac{2Ar_s}{4} \frac{\partial}{\partial r_C} \left( \frac{1}{r_C} \right) = -\frac{Ar_s}{2r_C^2} \\ \lambda' + \frac{2A^2}{r_C} &= -\frac{Ar_s}{2r_C^2} + \frac{2A^2}{r_C} \end{aligned}$$

These are not equal in general; so what went wrong? Why couldn't we use a regular transpose operation and get the same result? The key lies in the following conjecture: spacetime can not be extrinsically represented without a "partial operation."

Notice that our inner products throughout lecture 1 were all  $\langle u|v \rangle$ , the bra-ket combination. This meant that one argument was always conjugated; we never had a contraction without a conjugate tensor. This implicitly acted like a metric, flipping signs without regard for the arguments, and allowed us to construct  $\lambda$  such that the off-diagonal pieces vanished.

Conjugation acted as our partial operation, something that was "injected" into every contraction like the metric. In PXC, however, our partial operation (conjugation) was not dynamic- it wouldn't change with mass or time the way the metric does, because conjugation is only a "partial" operation; just a part that covers the information the metric encodes.

Pure Riemannian manifolds do not have this problem. Nash's embedding theorem tells us that any Riemannian manifold can be represented in an Euclidean space like the container. However, from my research, it seems that as soon as you step into a Lorentzian signature  $(+, +, +, -)$ , you lose the ability to represent the now pseudo-Riemannian manifold in an isotropic space. That's where the metric earns its place. It's also where the conjugation in the bra-ket becomes important, because without a partial operation you can't represent the Minkowski space.

Various theories give examples of partial operations. Regge calculus, teleparallel gravity, CDT; each one carries some injection of the Lorentzian signature that's unexplained beyond the axioms, the same way our bra-ket combination is unexplained beyond our axioms. Minkowski space is unavoidable to create full Relativity, and thus partial operations, or injected signatures, are as well.

PXC is named precisely because of this. We have not managed to embed a relativistic manifold in the container without a partial operation at all, but instead only with the help of conjugation in contractions. PXC is then another form of the same idea: representing spacetime through yet another partial operation. Whether you consider the metric, conjugation, Wick rotation, or any other partial operation "more natural" than any other is up to you.

Now that we've discussed a little of PXC's background and why conjugation is necessary, let's think about all the axioms used to replicate the Schwarzschild result. We have three things used:

1. Contractions move through conjugations.
2.  $\langle \dot{C} | \dot{C} \rangle_s = c^2$ .
3. Observable values are the real parts of complex values.

The first one is the standard partial operation definition, analogous to the statement that  $dx^i dx_i = g_{ij} dx^i dx^j$ . The second is a velocity normalization stating that everything inside the container moves at the same speed. Notice that for complex  $\dot{C}$ , this does not have to mean that an observable velocity is always present, but instead that some velocity (imaginary or real) must be. The third implies that really, PXC is a theory of seven dimensions; 2 for each observable spatial dimension and one time. The imaginary components are then auxiliary dimensions, unobservable but still interacting with our real three. Complexification is simply a method of packing all seven into four pieces.

For now, this is all for lecture 2; as we continue, should I see anything else that requires discussion, it will be added.